

GrabComponent

1. Component Overview

Purpose

GrabComponent is a VR object grabbing component attached to grabbable objects. It attaches objects precisely to the hand skeletal mesh through Sockets, and combines Control Rig-authored grab poses with the animation Blueprint blending system for natural VR hand grabbing. It supports multiple grab types (Free, Snap, Use Hand Socket, Custom) and physics-based throwing through the throw velocity function.

Design

- Socket-based precise grabbing: objects attach to predefined hand sockets rather than simply to a component
- Multi-socket support: the Use Socket Names array configures different sockets for left and right hands
- Control Rig pose authoring: the grab gesture for each object is authored in Sequencer with Control Rig
- Collision isolation: Pawn and PhysicsBody ignore each other in both directions to prevent VR physics conflicts

2. Core Variables

The following variables are defined in GrabComponent to control grabbing behavior and state:

Variable	Type	Description
Grab Type	Enum (E_GrabType)	Grab type: None / Free / Snap / Use Hand Socket / Custom
Use Socket Names	Array of Name	Array of socket names for both hands; Index 0 = right hand, Index 1 = left hand
Hand Grab Type	Enum (E_HandGrabType)	Grab pose enum: Default / HoldCube
Is Held	Boolean	Whether the object is currently held
Simulate on Drop	Boolean	Whether physics simulation is enabled on release
Motion Controller Ref	MotionControllerComponent	Motion controller reference
Hand	SkeletalMeshComponent	Hand skeletal mesh

Key points

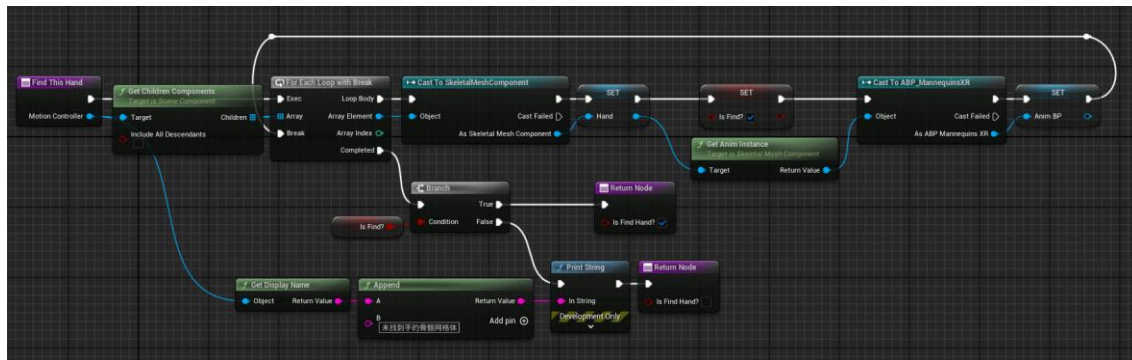
- Detach uses Keep World so the object keeps its world position and the drop position stays continuous
- After release, the Anim BP's Grab Type resets to Default, restoring the hand animation

5. Find This Hand Function

Description

Finds the hand skeletal mesh and animation Blueprint references among the MotionController's child components. Try Grab and Try Release obtain the hand references through this function.

Blueprint screenshots



Key points

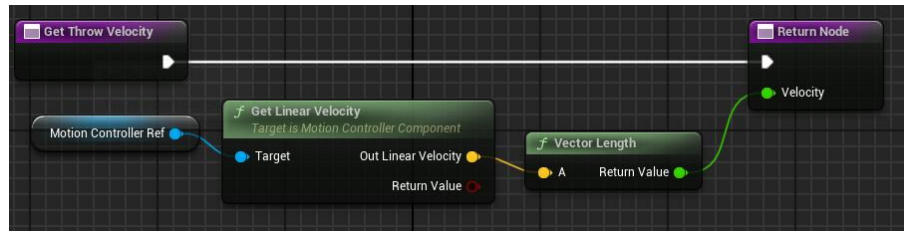
- Include All Descendants = false searches only direct children, because the hand mesh is always a direct child of the MC
- If the skeletal mesh's animation Blueprint is not ABP_MannequinsXR, no grab animation plays, but the hand reference is still valid
- For Each Loop with Break exits as soon as the first match is found, avoiding unnecessary iteration

6. Get Throw Velocity Function

Description

Gets the MotionController's current speed magnitude (scalar) for throw strength calculation.

Blueprint screenshots



Key points

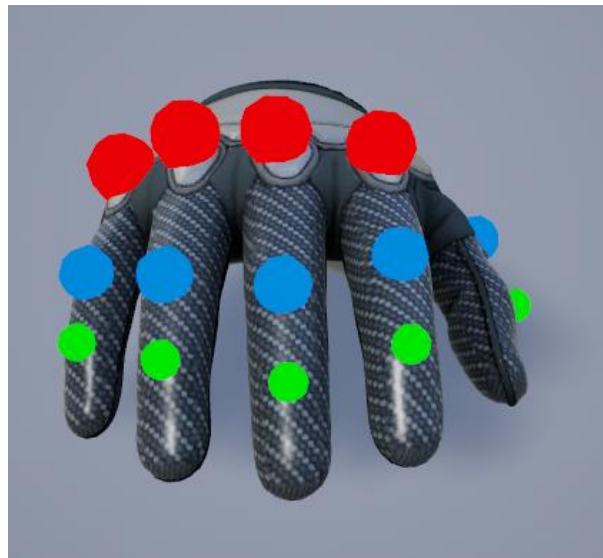
- Returns the speed magnitude (scalar) with no direction information
- Used to determine whether the player is throwing with a swing, and how strong the throw is

7. Control Rig Setup

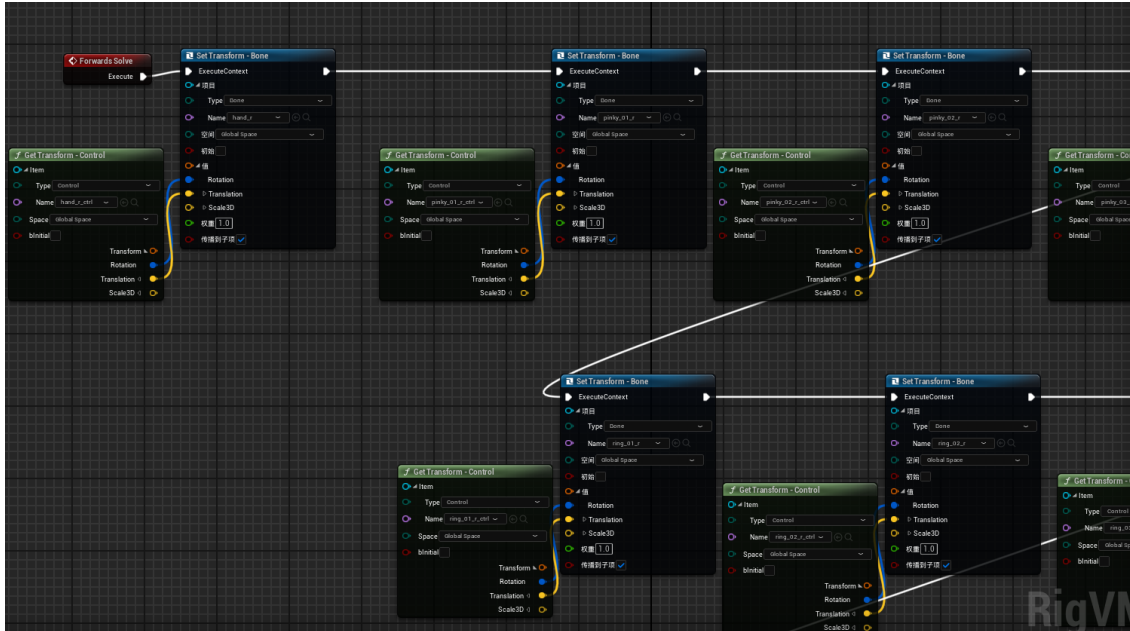
Description

Creates an FK Control Rig for the hand skeletal mesh so designers can pose the finger grab directly in Sequencer without pre-recorded animations.

Blueprint screenshot - control layout



Blueprint screenshot - Control Rig node graph



Steps

- In the skeletal mesh editor, add control points for each finger's three joints
 - Red = metacarpophalangeal joint, blue = proximal interphalangeal joint, green = distal interphalangeal joint

8. Socket Setup

Description

Adds Sockets to the hand skeletal mesh as the target positions for attaching objects to the hand. Sockets follow the bone animation so the object stays in the correct place while grabbed.

Steps

- Open the hand skeletal mesh editor;
- In the Skeleton Tree, right-click the target bone → Add Socket;
- Name the socket (e.g. S_Cube_R is the right-hand socket for grabbing the Cube);
- Adjust the socket's Relative Location / Rotation so the object sits correctly in the hand;
- Use Preview Asset to preview how the object aligns in the socket.



Key points

- Socket naming convention: S_ObjectName_Hand (e.g. S_Cube_R, S_Cube_L)
- Every grabbable object has a matching socket on the hand skeleton
- Sockets follow the bone animation so the object stays in the correct place while grabbed

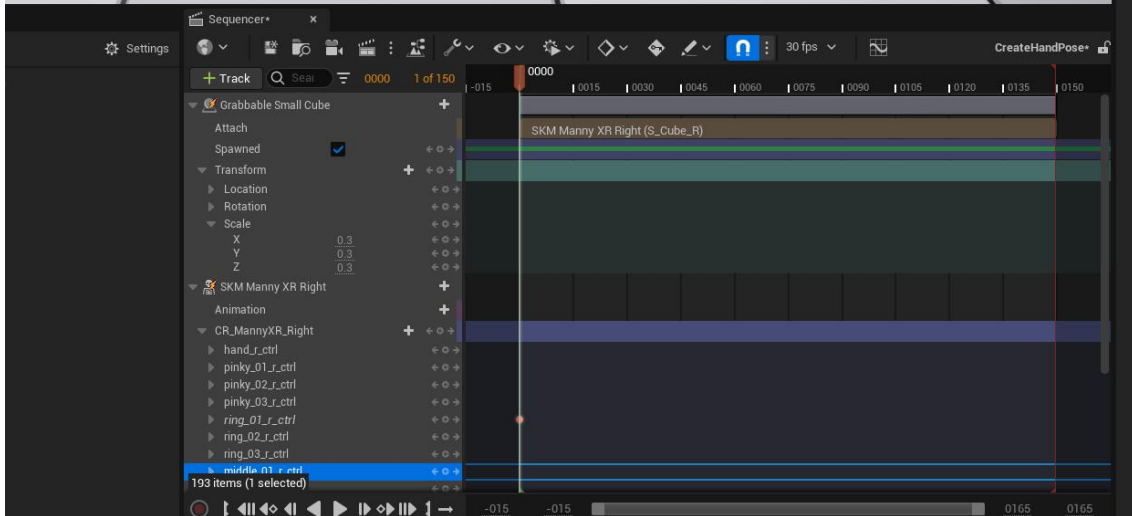
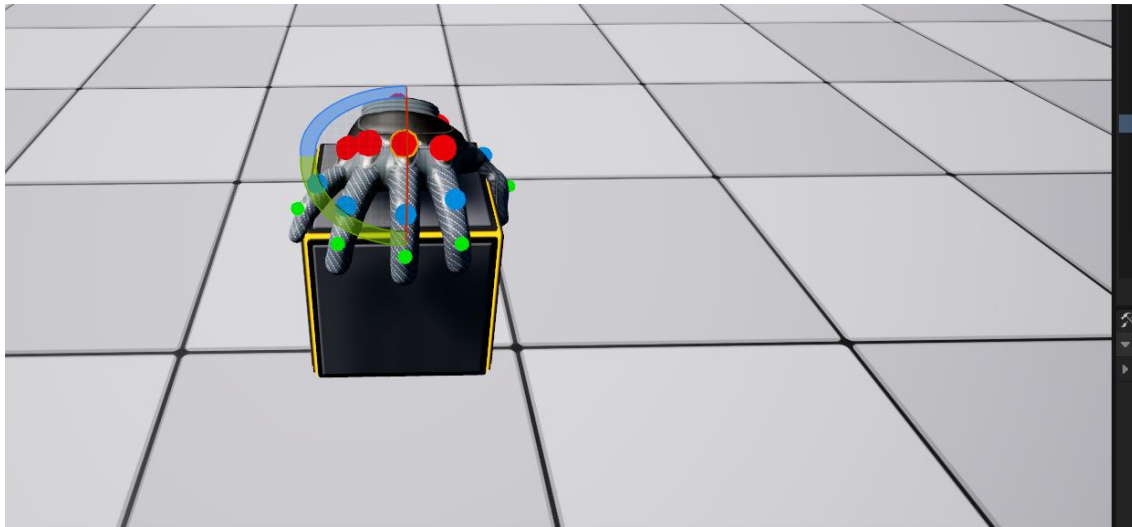
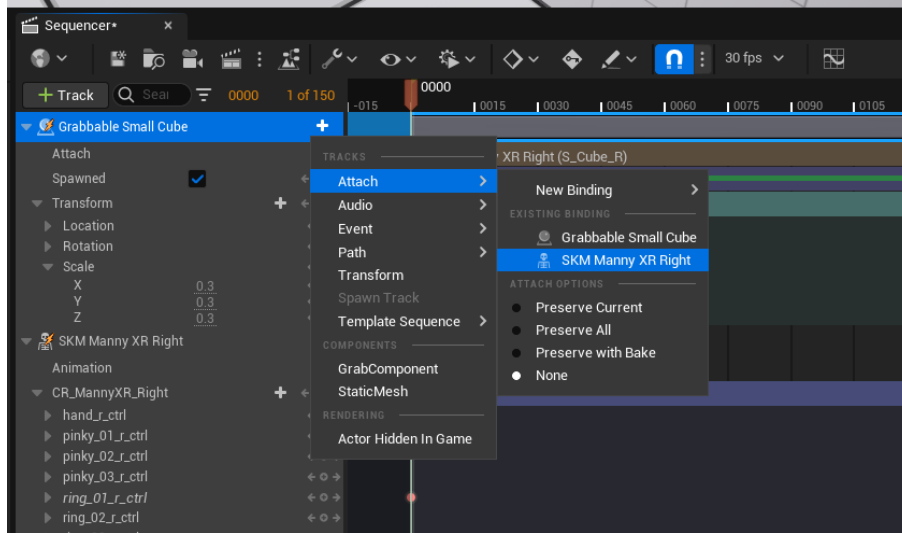
9. Grab Pose Authoring

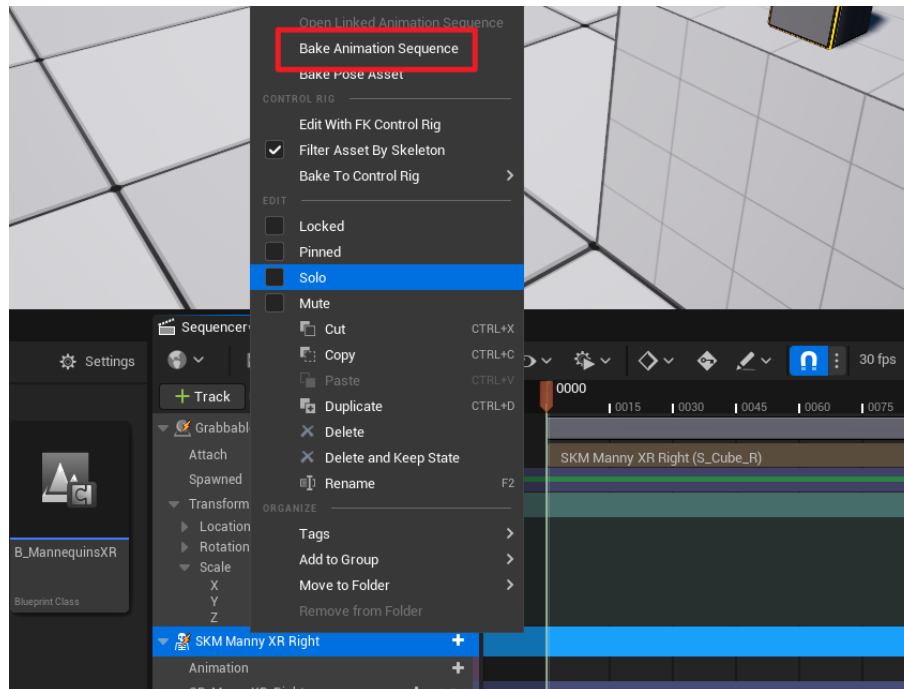
Description

Pose the grab gesture for a specific object in Sequencer with Control Rig, then export the keyframes as an animation asset (e.g. A_MannyXR_HoldCube) for the animation Blueprint to blend.

Steps

- Create a Level Sequence;
- Add the hand skeletal mesh (SKM_Manny XR Right);
- Add a grabbable object (e.g. Grabbable Small Cube, Scale 0.3) and attach it to the matching socket;
- In Sequencer, select the hand → + Animation → Control Rig → choose CR_MannyXR_Right
- Expand the finger control tracks and select/adjust each finger_XX_r_ctrl
- Export as an animation asset (e.g. A_MannyXR_HoldCube)





Key points

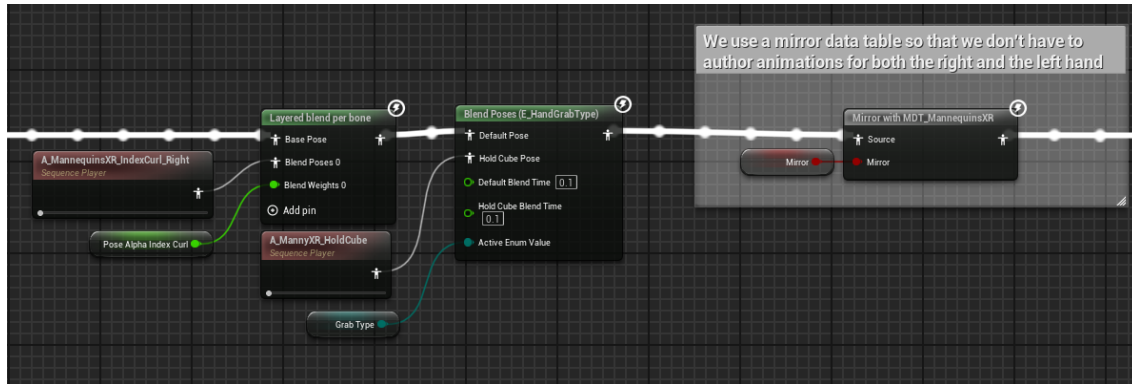
- Control Rig lets you pose directly in Sequencer without pre-recording animations
- Finger bends should wrap around the object's shape and avoid clipping through it
- Once the right-hand pose is done, the left-hand version is generated automatically through the Mirror Data Table
- Each object needs its own grab pose

10. Animation Blueprint Blending

Description

In ABP_MannequinsXR, the hand grab pose switches according to the Grab Type enum. Layered Blend overlays only the hand bones, Enum Blend switches poses, and Mirror reuses poses between hands.

Blueprint screenshots



Key points

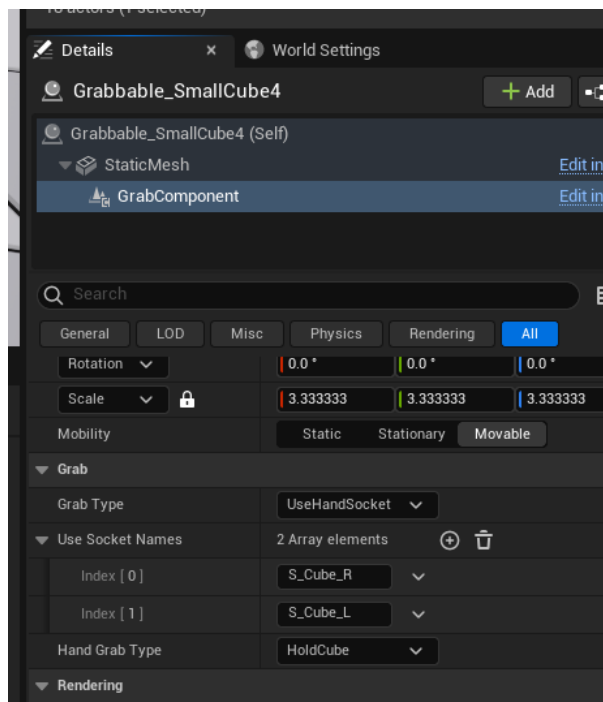
- Blend Poses by Enum selects the target pose by enum value; the 0.1-second transition avoids sudden jumps
- The Grab Type variable is set by GrabComponent in Try Grab / Try Release
- Only right-hand animations need to be authored; the left hand is generated by mirroring

11. GrabComponent Property Configuration

Description

Configure the GrabComponent's properties on the grabbable object's Blueprint or instance, specifying the grab method and matching sockets.

Blueprint screenshots

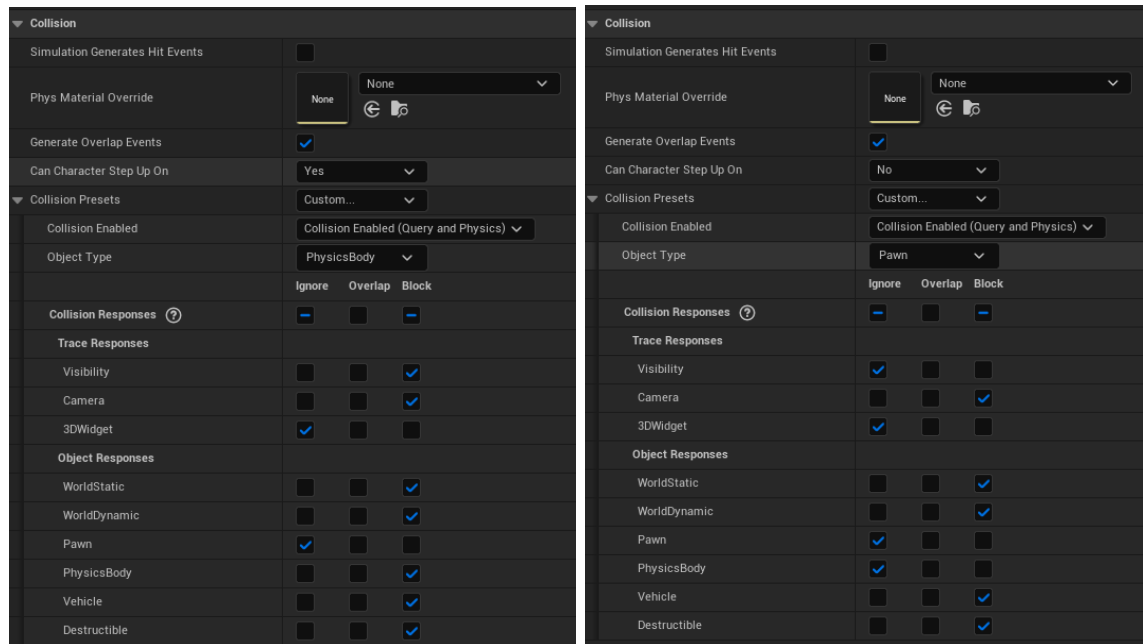


12. Collision Preset

Description

Modify the collision presets of the VR character and grabbable objects so the VR capsule collision does not disturb the grabbed object's physics. Since the VR character has capsule collision, restoring physics on release would cause conflicts if the two collided.

Blueprint screenshot - collision presets



Collision configuration

- Grabbable object (Cube): Object Type = PhysicsBody, Pawn = Ignore, Collision Enabled = Query and Physics
- VR Pawn: Object Type = Pawn, PhysicsBody = Ignore, Visibility = Overlap

Notes

- Pawn and PhysicsBody ignore each other both ways so released objects never physically conflict with the VR character
- After a released object regains physics, it cannot push the VR character
- The walking VR character cannot push physics objects on the ground
- Grabbable objects keep Block against WorldStatic / WorldDynamic so they never fall through the floor